AIRCRAFT PERFORMANCE – EXAMPLE PROBLEMS

Complete Solutions

PROBLEM 1: Glide Performance of Two Sailplanes

Theoretical Calculation:

Given:

- Wing reference area, $S = 15\ m^2$
- Drag polar: $C_D = 0.012 + 0.030\ C_L^2 \rightarrow C_{D0} = 0.012$, $K = 0.030$
- Release height, $h = 2{,}000$ m (calm air)
- Glider A: $W_A = 3{,}750$ N, $W_A/S = 250\ N/m^2$
- Glider B: $W_B = 3{,}750 + 1{,}500 = 5{,}250$ N, $W_B/S = 350\ N/m^2$
- Both fly at best glide angle (maximum L/D)
- Air density (not given – standard sea-level value assumed), $\rho = 1.225\ kg/m^3$

To find: (1) R_A, R_B (2) t_A, t_B

Step 1: Lift coefficient for best glide (maximum L/D)

$$C_L^* = \sqrt{(\ C_{D0} / K\ )}$$

where, C_L^* = lift coefficient at $(L/D)_{max}$, C_{D0} = zero-lift drag coefficient, K = induced-drag factor

$$C_L^* = \sqrt{\left(\frac{0.012}{0.030}\right)}$$

$$= \sqrt{0.4}$$

$$= 0.6325$$

Step 2: Drag coefficient at best glide

$$C_D^* = C_{D0} + K\,(C_L^*)^2 = 2\,C_{D0}$$

$$C_D^* = 0.012 + 0.030 \times (0.6325)^2 = 0.012 + 0.012$$

$$= 0.024$$

Step 3: Maximum lift-to-drag ratio

$$(L/D)_{max} = \frac{C_L^*}{C_D^*} = \frac{1}{2\sqrt{(C_{D0}\,K)}}$$

where, L = lift, D = drag

$$(L/D)_{max} = \frac{0.6325}{0.024} = 26.35$$

$$\text{Check: } = \frac{1}{2\sqrt{(0.012 \times 0.030)}} = \frac{1}{2 \times 0.01897} = 26.35$$

Step 4: Best glide angle

$$\tan\gamma_{min} = \frac{1}{(L/D)_{max}}$$

where, γ = glide (flight-path) angle below the horizontal

$$\tan\gamma_{min} = \frac{1}{26.35} = 0.03795$$

$$\gamma_{min} = \tan^{-1}(0.03795) = 2.173°$$

$$\sin\gamma = 0.03792\ ,\ \cos\gamma = 0.99928$$

Step 5: Maximum glide (ground) distance — Part 1

$$R = \frac{h}{\tan \gamma} = h\,(L/D)_{max}$$

where, R = horizontal distance covered, h = release height

$R = 2000 \times 26.35$

$R_A = R_B = 52{,}705 \text{ m} \approx 52.70 \text{ km}$

Result: $R_A = R_B \approx 52.7$ km. The best-glide distance depends only on $(L/D)_{max}$ (aerodynamics), not on weight; the ballast does not change how far the glider goes.

Step 6: Best-glide airspeed

$$V = \sqrt{\left(\frac{2\,(W/S)}{\rho\, C_L^*}\right)}$$

where, V = airspeed along the flight path, W/S = wing loading, ρ = air density

Glider A (W/S = 250 N/m²):

$$V_A = \sqrt{\left(\frac{2 \times 250}{1.225 \times 0.6325}\right)} = \sqrt{\left(\frac{500}{0.7748}\right)} = \sqrt{645.36} = 25.40 \text{ m/s}$$

Glider B (W/S = 350 N/m²):

$$V_B = \sqrt{\left(\frac{2 \times 350}{1.225 \times 0.6325}\right)} = \sqrt{\left(\frac{700}{0.7748}\right)} = \sqrt{903.51} = 30.06 \text{ m/s}$$

Step 7: Sink rate and time airborne — Part 2

$$V_v = V \sin \gamma$$

$$t = \frac{h}{V_v} = \frac{h}{V \sin \gamma}$$

where, V_v = rate of descent (sink speed), t = time to touchdown

Glider A:

$V_{v,A} = 25.40 \times 0.03792 = 0.963$ m/s

$$t_A = \frac{2000}{0.963} = 2076.2 \text{ s} \approx 34.60 \text{ min}$$

Glider B:

$V_{v,B} = 30.06 \times 0.03792 = 1.140$ m/s

$$t_B = \frac{2000}{1.140} = 1754.7 \text{ s} \approx 29.24 \text{ min}$$

Final answers – Problem 1

1) $R_A = R_B = 52.70$ km

2) $t_A = 2076$ s (34.6 min) ; $t_B = 1755$ s (29.2 min)

Glider B flies faster (30.1 vs 25.4 m/s), so it reaches the ground sooner but covers the same distance. ($t_A/t_B = \sqrt{(350/250)} = 1.183$)

PROBLEM 2: Cruise Range and Endurance of a Business Jet

Theoretical Calculation:

Given:

- W_0 = 100,000 N (start of cruise) ; W_1 = 80,000 N (fuel empty)
- Wing area, S = 30 m²
- Drag polar: $C_D = 0.016 + 0.045\, C_L^2 \rightarrow C_{D0} = 0.016$, K = 0.045
- Thrust specific fuel consumption (constant), $c_t = 0.65\ hr^{-1} = 1.805 \times 10^{-4}\ s^{-1}$
- Flight 1: sea level, $\rho_{SL} = 1.225\ kg/m^3$
- Flight 2: 33,000 ft, $\rho_{33k} = 0.380\ kg/m^3$
- Both flights trimmed for maximum range (jet, constant c_t)

To find: (1) R_{SL}, R_{33k} in km (2) E_{SL}, E_{33k}

Step 1: Lift coefficient for maximum range (jet aircraft)

For a jet the range is maximum when $\sqrt{C_L} / C_D$ is maximum, which gives:

$$C_{L,mr} = \sqrt{\left(\frac{C_{D0}}{3K}\right)}$$

where, $C_{L,mr}$ = lift coefficient for maximum range

$$C_{L,mr} = \sqrt{\left(\frac{0.016}{3 \times 0.045}\right)} = \sqrt{\left(\frac{0.016}{0.135}\right)} = \sqrt{0.11852} = 0.3443$$

Step 2: Drag coefficient and L/D at this condition

$$C_D = C_{D0} + K C_L^2 = C_{D0} + K \times \frac{C_{D0}}{3K} = (4/3)\, C_{D0}$$

$$C_D = 0.016 + 0.045 \times (0.3443)^2 = 0.016 + 0.005333 = 0.021333$$

$$L/D = \frac{C_L}{C_D} = \frac{0.3443}{0.021333} = 16.137$$

$$\sqrt{C_L} / C_D = \frac{\sqrt{0.3443}}{0.021333} = \frac{0.5867}{0.021333} = 27.503$$

Step 3: Range equation (Breguet, constant c_t, constant C_L)

$$R = \frac{2}{c_t}\sqrt{\left(\frac{2}{\rho S}\right)}\,\frac{\sqrt{C_L}}{C_D}\,(\sqrt{W_0} - \sqrt{W_1})$$

where, R = range, c_t = thrust specific fuel consumption (s^{-1}), ρ = air density, S = wing area, W_0, W_1 = initial and final weights

Common terms:

$$\sqrt{W_0} - \sqrt{W_1} = \sqrt{100000} - \sqrt{80000} = 316.228 - 282.843 = 33.385\ \sqrt{N}$$

$$2 / c_t = \frac{2}{1.805 \times 10^{-4}} = 11{,}080\ s$$

Flight 1 – Sea level (ρ = 1.225 kg/m³):

$$\sqrt{(2/\rho S)} = \sqrt{\left(\frac{2}{1.225 \times 30}\right)} = \sqrt{0.05442} = 0.23328$$

$$R_{SL} = 11{,}080 \times 0.23328 \times 27.503 \times 33.385$$

$$= 2{,}373{,}444\ m$$

$$R_{SL} = 2{,}373.4\ km$$

Flight 2 – 33,000 ft (ρ = 0.38 kg/m³):

$$\sqrt{(2/\rho S)} = \sqrt{\left(\frac{2}{0.38 \times 30}\right)} = \sqrt{0.17544} = 0.41885$$

$$R_{33k} = 11{,}080 \times 0.41885 \times 27.503 \times 33.385$$

$$= 4{,}261{,}429 \text{ m}$$

$$R_{33k} = 4{,}261.4 \text{ km}$$

Result: $R_{SL} \approx 2{,}373$ km ; $R_{33k} \approx 4{,}261$ km. Check: $R_{33k}/R_{SL} = \sqrt{(1.225/0.380)} = 1.795 = 1.795$ (the same L/D is used but the jet must fly faster in thin air, so it covers more ground per unit time).

Step 4: Endurance while flying the max-range profile

Fuel flow: $dW/dt = -c_t D = -c_t W / (L/D)$. With constant L/D (constant C_L), integrating gives:

$$E = \frac{1}{c_t}(L/D) \ln\left(\frac{W_0}{W_1}\right)$$

where, E = endurance (time airborne), L/D = lift-to-drag ratio at the flown condition

$$\ln\left(\frac{W_0}{W_1}\right) = \ln\left(\frac{100000}{80000}\right) = \ln(1.25) = 0.22314$$

$$E = \frac{1}{1.805 \times 10^{-4}} \times 16.137 \times 0.22314$$

$$= 5{,}540.2 \times 16.137 \times 0.22314$$

$$E = 19{,}950 \text{ s} = 5.54 \text{ hr}$$

E does not contain ρ, so it is the same at both altitudes:

$$E_{SL} = E_{33k} = 19{,}950 \text{ s} \approx 5.54 \text{ hr} (\approx 5 \text{ h } 32 \text{ min})$$

Final answers – Problem 2

1) $R_{SL} = 2{,}373.4$ km ; $R_{33k} = 4{,}261.4$ km

2) $E_{SL} = E_{33k} = 5.54$ hr (19,950 s)

Average speeds: R/E = 119 m/s at SL vs 214 m/s at 33,000 ft. Altitude increases range (higher speed for the same C_L) but leaves endurance unchanged.